

SOFTWARE ENGINEERING LAB – 2

Agile Backlog Creation & Sprint Simulation in Jira

Section C

Project	Faculty Research Grant & Publication Tracker
Jira Space / Board	SE Lab2 – Scrum / SLS board
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1. Objective

The objective of this lab is to create and manage a Scrum backlog in Jira, convert functional requirements into Epics and User Stories, estimate work using Story Points, plan and simulate sprints, move work through the Scrum workflow, and analyze the sprint using Jira reports.

2. Project Overview

The project used for the Jira backlog is the Faculty Research Grant & Publication Tracker. The backlog is organized around grant budget/expense management, publication management, and grant analytics/reporting.

3. Epics Created

Epic	Epic Name	Scope
E1	Grant Budget & Expense Management	Grant expense tracking, remaining budget, and spending-threshold escalation.
E2	Publication Management	Publication submission, duplicate checking, and co-author verification.
E3	Grant Analytics & Reporting	Grant fund burn-up analytics and reporting.

4. User Stories, Priorities and Story Points

The stories use the required format: “As a [user role], I want [goal], so that [benefit].” Fibonacci-style Story Points were used to represent relative effort and complexity.

ID	Epic	Summary	Priority	SP	User Story
SLS-4	Grant Budget & Expense Management	Track Grant Expenses and Budget Balance	High	8	As a Faculty Researcher, I want to log grant equipment and consumable expenses and view the remaining grant balance, so that I can monitor my available research funds.
SLS-5	Publication Management	Submit Publication and Check Duplicates	High	8	As a Faculty Researcher, I want to submit publication details and have the system check for duplicate DOI or title entries, so that publication records remain accurate and unique.
SLS-8	Publication Management	Verify Publication Through Co-author Approval	Medium	5	As a Faculty Researcher, I want co-authors to approve the authorship order and contribution share, so that a publication is verified only after all co-authors have approved it.
SLS-6	Grant Analytics & Reporting	View Grant Fund Burn-up Analytics	Medium	5	As a Research Dean, I want to view grant fund burn-up

					analytics filtered by grant, category and date range, so that I can monitor spending across research grants.
SLS-7	Grant Budget & Expense Management	Manage Grant Spending Threshold Escalation	High	8	As a Research Dean, I want to receive notifications when grant utilization crosses 75%, 90% or 100% and approve further spending, so that grant funds are not exceeded without authorization.

Total estimated backlog: 34 Story Points.

5. Sprint Planning and Simulation

Two Scrum sprints were simulated. Sprint 1 contained the grant expense tracking and publication-management stories. Sprint 2 contained the remaining analytics and grant-threshold escalation stories.

Sprint	Stories	Story Point Breakdown	Total SP
SLS Sprint 1	SLS-4, SLS-5, SLS-8	8 + 8 + 5	21
SLS Sprint 2	SLS-6, SLS-7	5 + 8	13

For the sprint simulation, work items were moved through the Scrum workflow from To Do to In Progress and then Done. The supplied Jira screenshots show Sprint 2 with SLS-6 and SLS-7 and a total of 13 Story Points.

6. Jira Evidence Screenshots

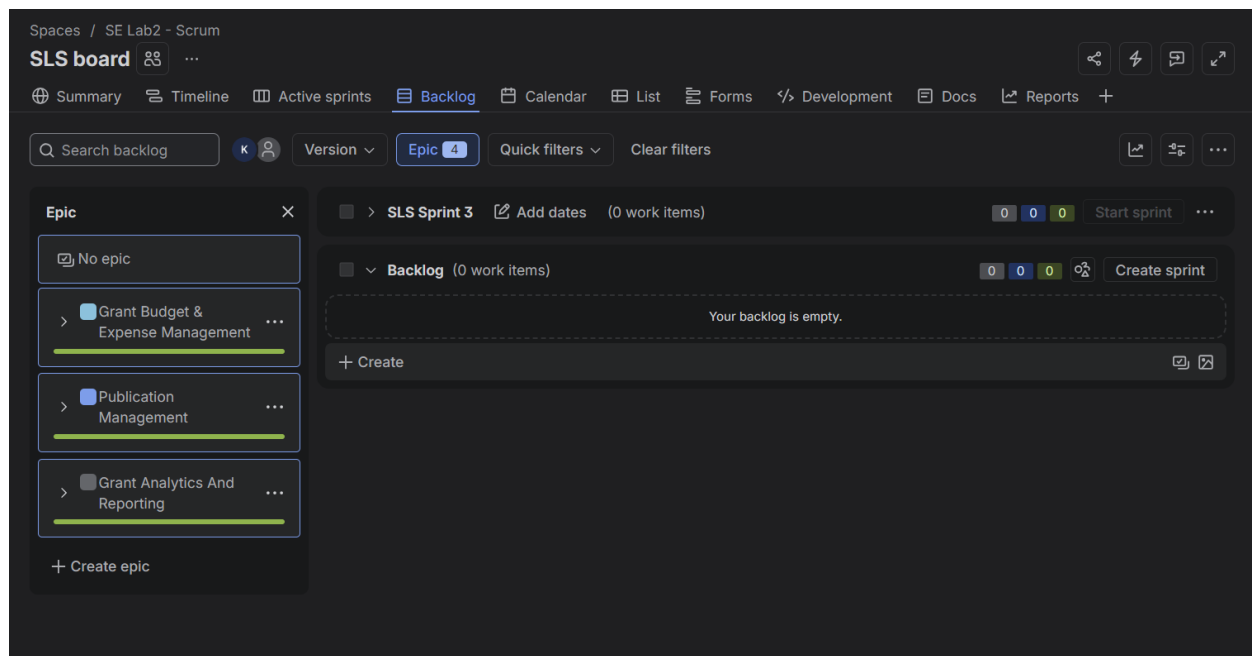


Figure 1. Jira Backlog showing the three project Epics.

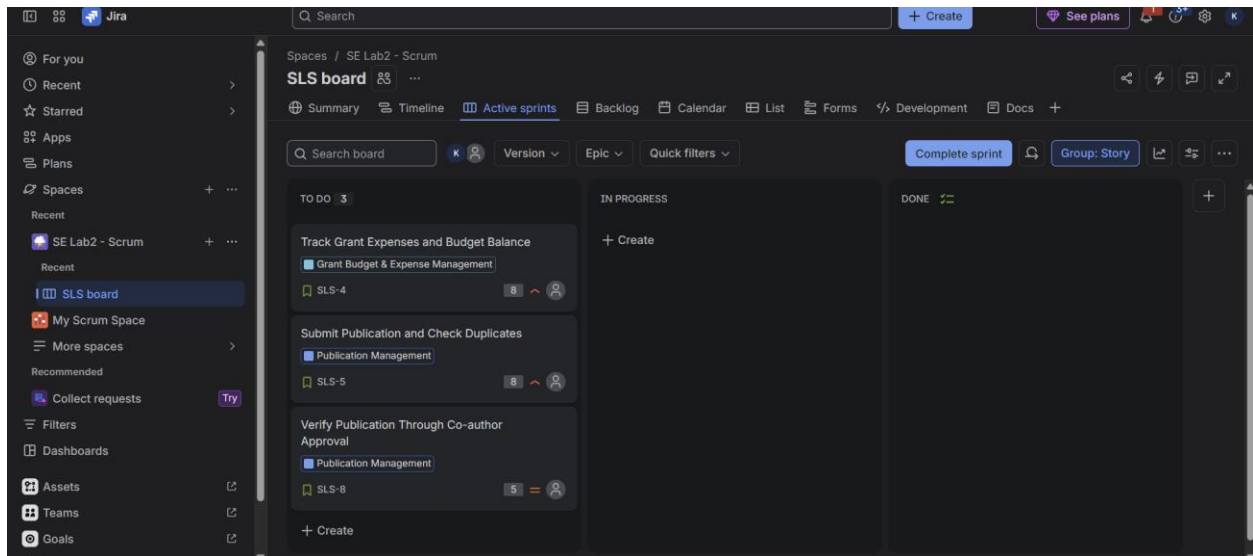


Figure 2. Jira Active Sprint board showing the Scrum workflow.

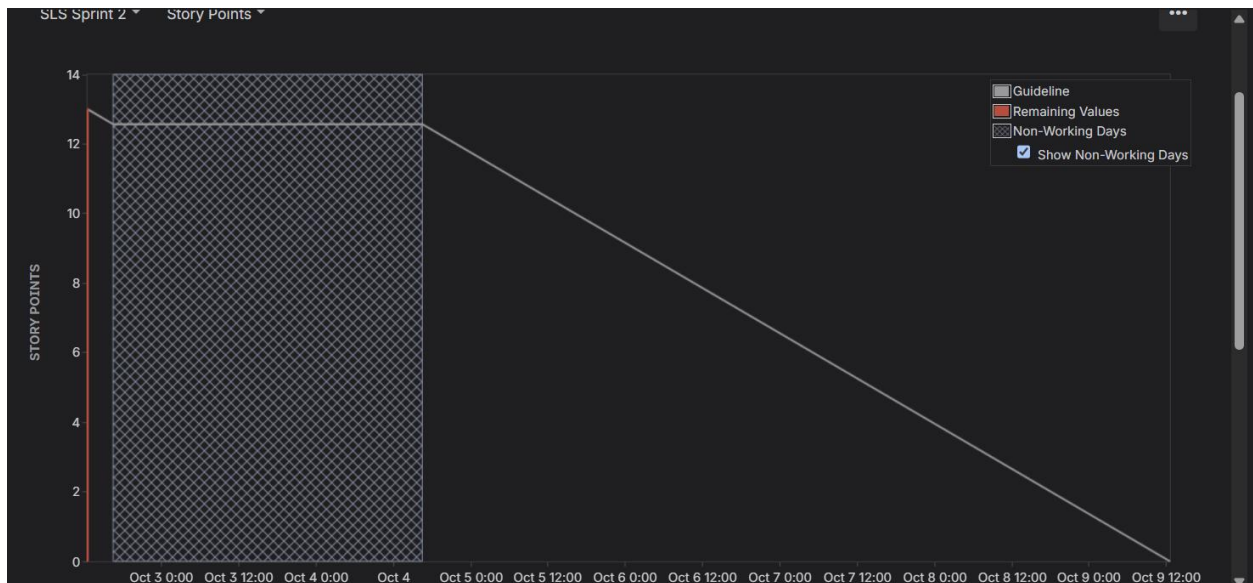


Figure 3. Jira Burndown Chart for SLS Sprint 2 using Story Points.

Date	Issue	Event Type	Event Detail	Inc.	Dec.	Remaining
02/Oct/26 12:32 PM	SLS-6	Sprint start		5		
	SLS-7			8		
	Total					13
02/Oct/26 12:33 PM	SLS-6	Burndown	Issue completed		5	8
	SLS-7	Burndown	Issue completed		8	0
	SLS-6	Sprint ended by keertikaacharya19				0
	SLS-7					0
	Total					0

Figure 4. Jira Sprint report/event details showing Sprint 2 start, completion of SLS-6 and SLS-7, and remaining Story Points reaching zero.

7. Burndown Chart Interpretation

The Burndown Chart is based on Story Points. Sprint 2 starts with 13 Story Points and the remaining-work line reaches 0 by the end of the simulated sprint. The chart therefore indicates that the planned Sprint 2 work was completed within the simulated sprint period.

8. Reflection

1. Did your estimations reflect the actual effort?

The estimates provided a reasonable relative representation of effort. The 8-point stories represented broader functionality, while the 5-point stories represented smaller but still non-trivial functionality. In the Sprint 2 simulation, the planned 13 Story Points were completed.

2. Was your backlog well-prioritized?

The backlog was prioritized using High and Medium priorities. Core grant budget and publication functions were treated as high-priority work, while analytics and co-author verification were assigned medium priority.

3. How did your simulated sprint align with your plan?

The Sprint 2 plan contained SLS-6 and SLS-7 for a total of 13 Story Points. The Jira sprint evidence shows both items being completed, so the simulated execution aligned with the planned Sprint 2 scope.

4. What insights did the burndown chart give about your team's capacity?

The chart shows that the team completed the full 13 Story Points planned for Sprint 2. This provides an initial indication of the team's simulated capacity. In a real project, several completed sprints would be needed before using velocity as a reliable capacity measure.

9. Conclusion

The Jira Scrum board was used to translate the Faculty Research Grant & Publication Tracker requirements into an actionable Agile backlog. The work was organized into Epics and User Stories, prioritized and estimated with Story Points, planned into sprints, executed through the Scrum board, and evaluated using the Burndown Chart. The Jira screenshots provide evidence of the backlog, sprint board, sprint completion and burndown results.